

E-commerce Return Rate Reduction Analysis

Internship Project Report

Project Objective

The aim of this project is to analyze an online retail dataset to understand customer purchasing behavior, identify products with a high return rate, and generate actionable business insights. The project combines Python-based data analysis with an interactive Power BI dashboard.

Dataset & Tools

Dataset: Online Retail Dataset.

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Power BI.

Project Workflow

1. Imported and explored the dataset.
2. Cleaned the data by removing duplicate records and rows with missing Customer IDs.
3. Created a TotalSales column using $\text{Quantity} \times \text{UnitPrice}$.
4. Performed Exploratory Data Analysis (EDA), including Top Countries by Sales, Monthly Sales Trend, Top Products, Top Customers, Sales Distribution, and Correlation Analysis.
5. Identified returned transactions using negative Quantity values and generated a High_Risk_Products.csv file.
6. Built a Logistic Regression model to classify return transactions and evaluated it using Accuracy Score, Confusion Matrix, and Classification Report.
7. Developed a Power BI dashboard containing KPIs, sales trends, product performance, and return analysis.

Key Findings

- The sales trend highlights seasonal business performance.
- A small number of products contribute significantly to total sales.
- Return analysis helps identify products that require quality improvement or better inventory management.
- The dashboard provides quick insights for business decision-making.

Conclusion

The project demonstrates the complete data analytics workflow—from data cleaning and visualization to predictive modeling and dashboard creation. The generated insights can help reduce product returns, improve customer satisfaction, and support informed business decisions.